

and overrules referee reports, and indirectly by influencing the referees that editors select. In the presence of strong policy and/or methodology preferences, a journal will tend to disproportionately publish papers that exhibit the editor's preferred papers. Such biases could have profound effects on the health and future of the discipline, given the large dependence of tenure decisions on T5 publications.

First, any biases have a direct effect on the composition of tenured scholars by decreasing the chances of publishing in the T5 for tenure-track faculty who do not cater to editorial preferences. This will cause access to the T5 to be more limited for scholars outside the network, which mechanically decreases their tenure rates.

Second, strong editorial preferences might also have an additional indirect effect by inducing future tenure-track faculty to only pursue those types of research that have been known to be published by an editor's journal. Pursuing this strategy is individual but socially irrational. Reliance on the T5 influences the course of future research.

6 Summary and Discussion

Without doubt, publication in the Top Five is a powerful determinant of tenure in academic economics that influences the choice of topics on which young economists work and squeezes papers into bite-sized journal-friendly fragments. One of us (Heckman) has had numerous conversations over the years with first-rate graduate students, post-doctoral fellows, and assistant professors about scientifically interesting research projects, only to be told

“that is a great idea, but it will not lead to a Top Five.”

An emphasis on publishing in the Top Five discourages large-scale, data-intensive empirical projects that explore and report the sensitivity of estimates to alternative assumptions. The fruits of such projects are often too long and do not easily fit into the format of the 40-page limit imposed by most of the Top Five journals.

Reliance on the T5 centralizes power to shape the profession into the hands of a select group of editors. Relying on the Top Five to screen the next generation of economists incentivizes professional incest and creates incentives for clientele effects whereby career-oriented authors appeal to the tastes of editors and the various biases of journals. It raises entry costs for new ideas and persons outside the orbits of the journals and their editors.⁷⁵

The current practice has weak empirical support if judged by its ability to produce papers that last in terms of citation counts. Publication in the Top Five is claimed to demonstrate the appeal of a paper to a broad base of professional economists assuming (without evidence) that subscribers read issues of journals cover to cover. The argument also ignores the fact that T5 referees are themselves field specialists, and field journals are highly influential outlets. Moreover, the Top Five journals do not have the highest impact factors even among economics journals, never mind general interest journals. Many non-T5 journals have citation counts that rival T5 journals, especially the lower-cited ones, such as *Review of Economic Studies* or *Econometrica*. Academics who impose the T5 standard impose a standard that they themselves do not follow. They primarily publish in, read and cite non-T5 journals, as will the candidates who survive the T5 filter and become tenured faculty.

Reliance on the T5 as a screening device raises serious concerns. First, an over-emphasis on T5 publications perversely incentivizes scholars to pursue follow-up and replication work at the expense of creative pioneering research since follow up work is easy to judge, is more likely to result in clean publishable results, and hence is more likely to be published.⁷⁶ This behavior is consistent with basic common sense: you get what you incentivize.

In light of the many adverse and potentially severe consequences associated with

⁷⁵Many readers of earlier drafts of this paper have remarked to us that the empirical results in this paper do not strictly prove these factors are operative. We grant this point. At the same time, we ask readers to apply the standard analysis of incentives to the “market” we have described. To deny the power and direction of these incentives is to assume an unlikely level of saintliness among journal editors and the scholars seeking to publish in their journals.

⁷⁶See the discussion at <https://www.aeaweb.org/webcasts/2017/curse>.

reliance on the T5, we believe it unwise for the discipline to continue using publication in the T5 as a measure of research achievement and as a predictor of future scholarly potential. The need for change is made ever more apparent by the T5's inadequacy as a predictor of individual article quality, much less the quality of a person. It also has an apparent gender tilt.

Our findings should spark a serious conversation in the profession about how to develop implementable alternatives to judge quality research. Such solutions would necessarily need to de-emphasize the role of the T5 in tenure and promotion decisions, and re-distribute the signalling function among more high-quality journals.⁷⁷ For example, there is limited evidence that *AEJ: Applied Economics* competes favorably with *Restat* and *EJ*.

However, a proper solution to the tyranny of the T5 will likely involve much more than a simple re-definition of the T5 to include a handful of additional influential journals. A better solution would address the flaw that is inherent in the practice of judging a scholar's potential for innovative work based on a track record of publications in a handful of journals selected by their impact factors.

In this issue, [Akerlof \(2018\)](#) sounds the alarm about the practice of relying on external rankings rather than individual reading of papers. The appropriate solution to the problem will require a significant shift from the current publications-based system of deciding tenure, to a system that emphasizes departmental peer-review of a candidate's work. Such a system would give serious consideration to unpublished working papers and the quality and integrity of a scholar's work. By closely reading published and unpublished papers rather than counting placements of publications, departments would signal that they both acknowledge and adequately account for the greater risk associated with serious scholars working at the frontiers of the discipline – an endeavor that is more likely to result in unpublished working papers chock-full of good ideas rather than T5 publications compared to other more

⁷⁷Due to their limited time in operation, we excluded the four new journals created by the American Economic Association: American Economic Journal: Microeconomics, American Economic Journal: Macroeconomics, American Economic Journal: Economic Policy, and American Economic Journal: Applied Economics

conventional and safer forms of research.⁷⁸

A more radical proposal is to shift publication away from the current fixed format journals and towards an open source arXiv or PLOS ONE format.⁷⁹ Such formats facilitate dissemination of new ideas and provide online realtime peer review for them. Discussion sessions would vet criticisms and provide both authors and their readers with different perspectives. Shorter, more focused papers would stimulate dialogue and break editorial and journal monopolies. An open source system would also allow authors to test new ideas in an arena of serious professional discussion and enable entry into the profession of creative out-of-network scholars. Networks and network-referential-citation circles are powerful barriers to entry into the profession that screen out new entrants with “odd ball” ideas and isolates those not acculturated in T5 values. Citing [Ellison \(2011\)](#) again, online publication is already being practiced by senior scholars. Why not broaden the practice and encourage spirited dialogue and rapid dissemination of new ideas?

Under any event, the profession should deemphasize crass careerism and promote creative activity. Short tenure clocks and reliance on the Top 5 to certify quality do just the opposite.

The importance of tolerating early failure and accounting for both the end-product and the path-to-production is illustrated in the analysis of [Manso \(2011\)](#), who studied optimal incentive-schemes for motivating innovation. Distinguishing between activities that *explore* new untested actions and those that *exploit* well-known actions, [Manso \(2011\)](#) shows that schemes aiming to promote *exploratory* activities should design reward structures to adjust for the higher variation associated with pay-offs from such activities. [Azoulay et al. \(2011\)](#) test this hypothesis on a sample of high-ability biomedical researchers by comparing

⁷⁸Some readers have objected that such a procedure would be too time-intensive and would require many to read out of their subfields. Others say that “quality” is a subjective thing. These objections go to the core of why economics departments exist if evaluating the work of a colleague is an onerous task. It is a symptom of departments that are collections of isolated scholars rather than a group that learns from fellow members.

⁷⁹See [Vale \(2015\)](#) for a discussion of the use of arXiv in Physics. See [Eisen \(2013\)](#) for remarks on PLOS ONE by Michael Eisen, its co-founder.

the publication outcomes of HHMI (Howard Hughes Medical Institute) grantees who enjoy more flexible and tolerant review processes with the publication outcomes of NIH (National Institute of Health) grantees who are subject to the “normal science” pre-defined deliverables, shorter review cycles and grant renewal policies that are unforgiving of failure. They find that, controlling for selection bias, HHMI grantees published high-impact articles at a higher rate than NIH grantees. More importantly, HHMI grantees appeared more likely to engage in *exploratory* research, as suggested by a lower degree of overlap between the MeSH (Medical Subject Headers) keywords associated with works published during the pre- and post-grant periods.

In the long run, the profession will benefit from application of more creativity-sensitive screening of its next generation. Otherwise, academic economics risks becoming (or remaining) a group of Top Five plodders putting one foot in front of the other. Emphasis on the T5 in sorting talent creates a culture where vitae length and publication speed in select journals rather than the development of a body of coherent and original ideas is most valued. It incentivizes careerism rather than creative scholarship.